

ANSHUL VASMATE

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EDUCATION

University of Maryland, College Park — B.S. Computer Engineering Expected May 2030
Coursework: *Object-Oriented Programming I (Java), Calculus II, Intro to Engineering Design (CAD), General Chemistry*

James M. Bennett High School, Salisbury, MD 2022 – 2026
AP Scholar (2025) · National Honor Society · Mu Alpha Theta · Science National Honor Society

TECHNICAL PROJECTS

ShortsForge — AI Video Generation Pipeline / Python, Ollama (Llama 3.1), Anthropic API, Kokoro TTS, MoviePy 2026

- Built a Python CLI that turns one topic into a publish-ready YouTube Short: script, scene plan, image prompts, voiceover, captioned video and metadata.
- Ran generation on a local LLM and local text-to-speech (zero API cost), with swappable backends for the Anthropic API and cloud voices.
- Designed two AI agents that run the pipeline stage by stage with human approval gates; kept AI characters visually consistent with a single TOML source of truth; produced 3 Shorts and a 10+ minute long-form script.

Machine-Learning Stock-Ranking Research Engine / Python, pandas, scikit-learn, FastAPI, Alpaca API 2026

- Built a backtesting pipeline free of look-ahead bias over 374K rows (128 stocks, 2015–2026) and tested 2,328 model/portfolio configurations with walk-forward validation.
- The selected model held up on a quarantined holdout (IC +0.024, $t = 2.39$) and beat all 40 random-ranking controls ($p \leq 0.024$).
- Wrote a 38-assertion test suite to catch data leakage and built a paper-trading harness to measure real fill costs.

MotionVision — Hand-Gesture Computer Control / Python, OpenCV, MediaPipe, pytest 2026

- Built a desktop app that uses real-time webcam hand tracking to control the mouse cursor and trigger user-recorded gestures (keystrokes, media, app launches).
- Smoothed cursor jitter with a One Euro filter; fixed Windows DPI scaling, multi-monitor, and stuck-drag bugs; covered the fixes with a 29-test pytest regression suite.

AI Agent Tooling — 5 Model Context Protocol (MCP) Servers / Python, FastMCP, Pydantic, Windows UI Automation 2026

- Built MCP servers that let AI assistants control desktop apps, a job-search tracker, a task planner, and a personal knowledge base in plain English (30+ tools).
- Built safety into every tool: allowlist-only app launching, path-traversal protection, and a desktop-control server with 5 safety gates, rate limiting, and an audit log. Verified with 130+ automated test assertions.

Voice AI Assistant & Windows Productivity Suite / Python, pywebview, Tkinter, PowerShell, SpeechRecognition 2026

- Built a voice-controlled daily dashboard (weather, tasks, 99 synced class deadlines) running a local LLM that manages the user's planner through speech.
- Shipped 7 Windows tray utilities (clipboard history, launcher with global hotkeys, focus timer, and others). Found why the apps silently failed to start: stale OS-level lock files.

LEADERSHIP & ACTIVITIES

VEX V5 Robotics Team — Team Captain & Lead Programmer 2022 – 2026

- Led a 5–6 member team and programmed the robot in Python, qualifying for the DelMarVa Regional Championship (4th); mentored new members and helped Salisbury University's team build a prototype.

AI Prompt Generator — Independent Student Researcher 2024 – 2026

- Explored LangChain and ran 100+ experiments on AI-assisted studying; built and released a prompt-generator tool (5+ community downloads).

UMES Robotic-Arm Research — Research Participant 2025 – 2026

- Worked on a machine-learning robotic-arm project; trained TensorFlow on 600+ images under guidance from UMES students and a professor.

Future Health Professionals Club — Co-Founder & Vice President 2025 – 2026

- Grew the club to 60+ members; brought in guest speakers and set up a university mentorship partnership.

Stock Market Club (Treasurer) · **Youth Environmental Action** (Web Developer) · **ENgin** (Mentor) 2022 – 2026

- Managed club funds and led a 20-student university trip; built the YEA/TidalHealth community website; tutored Ukrainian students in English for 30+ hours.

AWARDS

- 1st Place**, JA Eastern Shore Stock Market Challenge (2025) · **1st Place**, Science Olympiad Geological Mapping · **Bayside Conference Champion**, Varsity Tennis

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML/CSS, PowerShell

AI / ML: LLM agents, MCP, Claude Code, Anthropic API, Ollama, LangChain, scikit-learn, pandas, NumPy, OpenCV, MediaPipe

Tools: Git/GitHub, FastAPI, pytest, Linux (WSL), Onshape CAD, Windows UI Automation